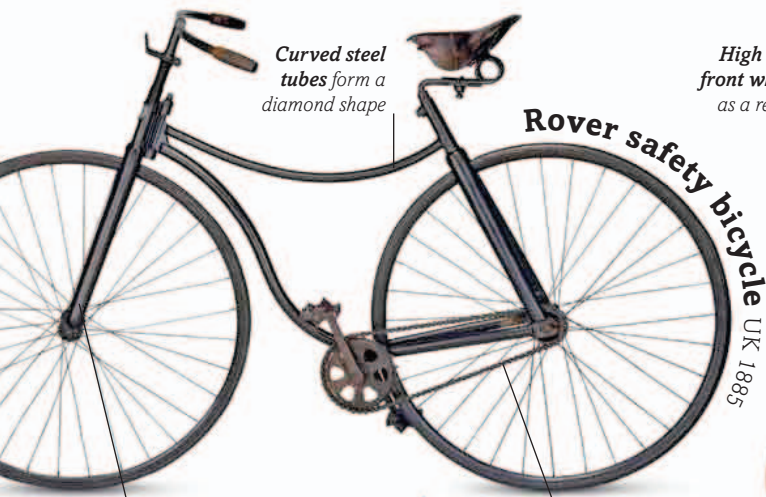




High wheeler front wheel acts as a rear wheel

Straight front forks

Large front wheel up to 5 ft (1.5 m) in diameter

In **1885**, the Rover bicycle won a **100 mile (160 km)** race in the UK in 7 hours, 5 minutes.



Singer tricycle UK 1888

Solid rubber tire, replaced by pneumatic (air-filled) tires from 1888



Swift safety bicycle UK 1887

Simple mudguard covered the small rear wheel



Facile dwarf safety bicycle UK 1888



Curved front forks

Singer safety UK 1888

In **1884**, Thomas Stevens crossed the **USA** on a **high wheeler**.



Saddle made of 131 ft (40 m) of woven cord weighs 3½ oz (100 g)

Dursley Pedersen bicycle UK 1898-99

to the front wheel. The experience of its iron “tires” on cobbled streets earned it the nickname *boneshaker*. **High wheelers**, or Penny Farthings, in the UK, France, and the USA had no chains or gears, but had bigger front wheels to boost speed. It perched the rider high above the ground,

resulting in many falls. Alternatives were sought, including pairing two high wheeler front wheels to form the rear wheels of the **Singer tricycle**, and using a chain-driven rear wheel, as in the **Rover safety bicycle**. This design ushered in the modern bicycle with wheels of similar size.